

Simplex method

5

Maximize:

$$Z = 3x_1 + 2x_2$$

Subject to:

$$x_1 + x_2 \leq 4$$

$$2x_1 + x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

Solution

Convert inequalities to equations by adding slack variables

Introduce slack variables s_1 and s_2 to turn the inequalities into equalities.

Step 1: Convert inequalities to equations by adding slack variable

Introduce slack variables s_1 and s_2 to turn the inequalities into equalities.

$$1. \ x_1 + x_2 + s_1 = 4$$

$$2. \ 2x_1 + x_2 + s_2 = 5$$

The objective function in standard form becomes:

$$Z - 3x_1 - 2x_2 = 0$$

Adding the slack variables s_1 and s_2 with zero coefficients:

$$Z - 3x_1 - 2x_2 + 0s_1 + 0s_2 = 0$$

Iteration 1

Construct the initial simplex tableau

Basis	x_1	x_2	s_1	s_2	RHS
s_1	1	1	1	0	4
s_2	2	1	0	1	5
Z	-3	-2	0	0	0

Solve using the simplex method

1. Identify the entering variable:

The most negative coefficient in the Z-row is -3 (corresponding to x_1). Thus, x_1 enters the basis.

2. **Determine the leaving variable:**

Compute the ratio of RHS to the pivot column (x_1) to determine the leaving variable:

- For s_1 : $4/1=4$
- For s_2 : $2/2=2.5$

The smallest ratio is 2.5, so s_2 leaves the basis.

1. **Perform the pivot operation:**

Normalize the pivot row (row 2) and update the tableau. Let me calculate this.

2. $R_2 = R_2 / 2$

3. $R_1 = R_1 - R_2$

4. $R_3 = R_3 + 3R_2$

After the first pivot operation and Row transformation, the updated simplex tableau is:

Basis	x_1	x_2	s_1	s_2	RHS
s_1	0	0.5	1	-0.5	1.5
x_1	1	0.5	0	0.5	2.5
Z	0	-0.5	0	1.5	7.5

Continue the simplex method

1. **Identify the entering variable:**

The most negative coefficient in the Z row is -0.5 (corresponding to x_2). Thus, x_2 enters the basis.

2. **Determine the leaving variable:**

Compute the ratio of RHS to the pivot column (x_2):

- For s_1 : $1.5/0.5=3$
- For x_1 : $2.5/0.5=5$

The smallest ratio is 3, so s_1 leaves the basis.

Perform the Row transformations

$R_1 = R_1 / 0.5$

$R_2 = R_2 - R_1$

$R_3 = R_3 + R_1$

After the second pivot operation, the updated simplex tableau is:

Basis	x_1	x_2	s_1	s_2	RHS
x_2	0	1	2	-1	3
x_1	1	0	-1	1	1
Z	0	0	1	1	9

There are no negative coefficients in the Z -row, indicating that the solution is optimal.

Final Solution:

- $x_1=1$
- $x_2=3$
- Slack variables: $s_1=0, s_2=0$
- Maximum value of $Z=9$

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Maximize:

$$Z = 3x_1 + 2x_2$$

Subject to:

$$x_1 + x_2 \leq 4$$

$$x_1 \leq 5$$

$$x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

Convert inequalities to equations

Introduce slack variables s_1, s_2, s_3 for the inequalities to rewrite them as equalities:

1. $x_1 + x_2 + s_1 = 4$
2. $x_1 + s_2 = 5$
3. $x_2 + s_3 = 6$

The objective function in standard form is:

$$Z - 3x_1 + 2x_2 - 0s_1 - 0s_2 - 0s_3 = 0$$

Iteration 1

Basis	x_1	x_2	s_1	s_2	s_3	RHS
s_1	1	1	1	0	0	4
s_2	1	0	0	1	0	5
s_3	0	1	0	0	1	6
Z	-3	-2	0	0	0	0

Solve using the simplex method**1. Entering variable:**

The most negative coefficient in the Z row is -3 (corresponding to x_1). So, x_1 enters the basis.

2. Leaving variable:

Compute the ratio of RHS to the pivot column (x_1):

- For s_1 : $4/1=4$
- For s_2 : $5/1=5$
- For s_3 : Not applicable (x_1 's coefficient is 0).

The smallest ratio is 4, so s_1 leaves the basis.

Perform the pivot operation:

Normalize the pivot row and update the tableau. Let me compute this.

Pivot element is already 1. Perform the following operations

$$R_2 = R_2 - R_1$$

R_3 , x_1 is already 0, no transformation is required for row 3

$$R_4 = R_4 + 3R_1$$

After the first pivot operation, the updated simplex tableau is:

Basis	x_1	x_2	s_1	s_2	s_3	RHS
x_1	1	1	1	0	0	4
s_2	0	-1	-1	1	0	1
s_3	0	1	0	0	1	6
Z	0	1	3	0	0	12

Check for optimality

The Z -row has no negative coefficients, meaning the current solution is optimal.

Final Solution:

- **Basic Variables:**
 - $x_1 = 4$
 - $x_2 = 0$
 - Slack variables: $s_1 = 0, s_2 = 1, s_3 = 6$
- **Maximum Value of Z : $Z = 12$**

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Find the minimum using gradient descent $f(x) = x^2 - 6x + 9$ from the point $x = 0$, taking the learning rate 0.1. Perform 10 iterations.

Solution

$$f'(x) = \frac{d}{dx}(x^2 - 6x + 9) = 2x - 6$$

	<table> <tr> <th>Iteration</th><th>x_t</th></tr> <tr><td>0</td><td>0.0000</td></tr> <tr><td>1</td><td>0.6000</td></tr> <tr><td>2</td><td>1.0800</td></tr> <tr><td>3</td><td>1.4640</td></tr> <tr><td>4</td><td>1.7712</td></tr> <tr><td>5</td><td>2.01696</td></tr> <tr><td>6</td><td>2.21357</td></tr> <tr><td>7</td><td>2.37085</td></tr> <tr><td>8</td><td>2.49668</td></tr> <tr><td>9</td><td>2.59735</td></tr> <tr> <td>10</td><td>2.67788</td></tr> </table>	Iteration	x_t	0	0.0000	1	0.6000	2	1.0800	3	1.4640	4	1.7712	5	2.01696	6	2.21357	7	2.37085	8	2.49668	9	2.59735	10	2.67788
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9	Minimize the function $(x)=x^2+2y^2$ subject to the constraint $x+y=4$. Solution Let's minimize the function $f(x, y) = x^2 + 2y^2$ using the given Lagrangian: $\mathcal{L}(x, y, \lambda) = x^2 + 2y^2 - \lambda(x + y - 4)$																								

Step 1: Compute partial derivatives

We take the partial derivatives of $\mathcal{L}$ with respect to x , y , and λ :

1. $\frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda = 0$
2. $\frac{\partial \mathcal{L}}{\partial y} = 4y - \lambda = 0$
3. $\frac{\partial \mathcal{L}}{\partial \lambda} = -(x + y - 4) = 0$

Step 2: Solve the system of equations

From the first equation:

$$\lambda = 2x$$

From the second equation:

$$\lambda = 4y$$

Equating $2x = 4y$:

$$x = 2y$$

From the third equation:

$$x + y = 4$$

Substitute $x = 2y$ into $x + y = 4$:

$$2y + y = 4 \implies 3y = 4 \implies y = \frac{4}{3}$$

Substitute $y = \frac{4}{3}$ into $x = 2y$:

$$x = 2 \cdot \frac{4}{3} = \frac{8}{3}$$

Step 3: Compute the minimum value of $f(x, y)$

Substitute $x = \frac{8}{3}$ and $y = \frac{4}{3}$ into $f(x, y) = x^2 + 2y^2$:

$$\begin{aligned} f\left(\frac{8}{3}, \frac{4}{3}\right) &= \left(\frac{8}{3}\right)^2 + 2\left(\frac{4}{3}\right)^2 \\ f\left(\frac{8}{3}, \frac{4}{3}\right) &= \frac{64}{9} + 2 \cdot \frac{16}{9} = \frac{64}{9} + \frac{32}{9} = \frac{96}{9} = \frac{32}{3} \end{aligned}$$

1
0

Minimize $f(x,y)=2x+3y$, subject to the constraint, $g(x,y)=x^2+y^2-9=0$

Solution

$$\mathcal{L}(x, y, \lambda) = 2x + 3y - \lambda(x^2 + y^2 - 9)$$

We aim to minimize $f(x, y) = 2x + 3y$ subject to the constraint $g(x, y) = x^2 + y^2 - 9 = 0$.

Step 1: Compute Partial Derivatives

We compute the partial derivatives of $\mathcal{L}(x, y, \lambda)$ with respect to x , y , and λ , and set them to zero:

1. $\frac{\partial \mathcal{L}}{\partial x} = 2 - 2\lambda x = 0$
2. $\frac{\partial \mathcal{L}}{\partial y} = 3 - 2\lambda y = 0$
3. $\frac{\partial \mathcal{L}}{\partial \lambda} = -(x^2 + y^2 - 9) = 0$

Step 2: Solve the System of Equations

From the first equation:

$$2 = 2\lambda x \implies \lambda = \frac{1}{x} \quad (x \neq 0)$$

From the second equation:

$$3 = 2\lambda y \implies \lambda = \frac{3}{2y} \quad (y \neq 0)$$

Equating λ from the two equations:

$$\frac{1}{x} = \frac{3}{2y} \implies 2y = 3x \implies y = \frac{3}{2}x$$

From the constraint:

$$x^2 + y^2 = 9$$

Substitute $y = \frac{3}{2}x$ into the constraint:

$$x^2 + \left(\frac{3}{2}x\right)^2 = 9$$

$$x^2 + \frac{9}{4}x^2 = 9$$

$$\frac{13}{4}x^2 = 9 \implies x^2 = \frac{36}{13} \implies x = \pm \frac{6}{\sqrt{13}}$$

Solve for y :

Substitute $x = \pm \frac{6}{\sqrt{13}}$ into $y = \frac{3}{2}x$:

$$y = \pm \frac{3}{2} \cdot \frac{6}{\sqrt{13}} = \pm \frac{9}{\sqrt{13}}$$

The two critical points are:

$$(x, y) = \left(\frac{6}{\sqrt{13}}, \frac{9}{\sqrt{13}}\right) \quad \text{and} \quad \left(-\frac{6}{\sqrt{13}}, -\frac{9}{\sqrt{13}}\right)$$

Step 3: Evaluate $f(x, y)$ at Critical Points

At $\left(\frac{6}{\sqrt{13}}, \frac{9}{\sqrt{13}}\right)$:

$$f\left(\frac{6}{\sqrt{13}}, \frac{9}{\sqrt{13}}\right) = 2\left(\frac{6}{\sqrt{13}}\right) + 3\left(\frac{9}{\sqrt{13}}\right)$$

$$f\left(\frac{6}{\sqrt{13}}, \frac{9}{\sqrt{13}}\right) = \frac{12}{\sqrt{13}} + \frac{27}{\sqrt{13}} = \frac{39}{\sqrt{13}}$$

At $\left(-\frac{6}{\sqrt{13}}, -\frac{9}{\sqrt{13}}\right)$:

$$f\left(-\frac{6}{\sqrt{13}}, -\frac{9}{\sqrt{13}}\right) = 2\left(-\frac{6}{\sqrt{13}}\right) + 3\left(-\frac{9}{\sqrt{13}}\right)$$

$$f\left(-\frac{6}{\sqrt{13}}, -\frac{9}{\sqrt{13}}\right) = -\frac{12}{\sqrt{13}} - \frac{27}{\sqrt{13}} = -\frac{39}{\sqrt{13}}$$

Final Answer:

The minimum value of $f(x, y)$ is:

$$\boxed{-\frac{39}{\sqrt{13}}}$$

This occurs at:

$$(x, y) = \left(-\frac{6}{\sqrt{13}}, -\frac{9}{\sqrt{13}}\right)$$

1

1

Minimize $f(x, y) = x^2 + y^2$, subject to the constraint $g(x, y) = x + y - 1 \leq 0$.

Solution

$$\mathcal{L}(x, y, \lambda) = x^2 + y^2 - \lambda(x + y - 1)$$

Step 1: Compute Partial Derivatives

To find the critical points, compute the partial derivatives of $\mathcal{L}(x, y, \lambda)$ with respect to x , y , and λ , and set them to zero:

$$1. \quad \frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda = 0 \quad \implies \lambda = 2x$$

$$2. \quad \frac{\partial \mathcal{L}}{\partial y} = 2y - \lambda = 0 \quad \implies \lambda = 2y$$

$$3. \quad \frac{\partial \mathcal{L}}{\partial \lambda} = -(x + y - 1) = 0 \quad \implies x + y = 1$$

Step 2: Solve the System of Equations

From (1) and (2):

$$2x = 2y \implies x = y$$

Substitute $x = y$ into the constraint (3):

$$x + y = 1 \implies x + x = 1 \implies 2x = 1 \implies x = \frac{1}{2}$$

Thus, $x = \frac{1}{2}$ and $y = \frac{1}{2}$.

Step 3: Evaluate the Objective Function

Substitute $x = \frac{1}{2}$ and $y = \frac{1}{2}$ into $f(x, y) = x^2 + y^2$:

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Answer

The **minimum value** of $f(x, y) = x^2 + y^2$ is:

$$\boxed{\frac{1}{2}}$$

This occurs at:

$$\boxed{x = \frac{1}{2}, y = \frac{1}{2}}$$

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2

Solution

We aim to **maximize** $f(x, y) = 3x + 4y$ subject to the constraint $g(x, y) = x^2 + y^2 - 4 \leq 0$.

This is a constrained optimization problem, and since it involves an inequality constraint, we will apply the **Karush-Kuhn-Tucker (KKT) conditions**.

Step 1: Lagrangian Formulation

Define the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = 3x + 4y + \lambda(4 - x^2 - y^2)$$

where $\lambda \geq 0$ is the Lagrange multiplier for the inequality constraint.

Step 2: KKT Conditions

The KKT conditions include:

1. **Stationarity:** The gradient of the Lagrangian must vanish:

$$\frac{\partial \mathcal{L}}{\partial x} = 3 - 2\lambda x = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial y} = 4 - 2\lambda y = 0 \quad (2)$$

2. **Primal feasibility:** The constraint must hold:

$$x^2 + y^2 - 4 \leq 0 \quad (3)$$

3. **Dual feasibility:** $\lambda \geq 0$.

4. **Complementary slackness:**

$$\lambda(x^2 + y^2 - 4) = 0 \quad (4)$$

Step 3: Solve the System of Equations

Case 1: The constraint is not active ($x^2 + y^2 - 4 < 0$)

If the constraint is not active, $\lambda = 0$. From (1) and (2):

$$3 = 0 \quad (\text{not possible})$$

Thus, the constraint must be active.

Case 2: The constraint is active ($x^2 + y^2 - 4 = 0$)

If the constraint is active, $\lambda \neq 0$. From (1):

$$3 = 2\lambda x \implies \lambda = \frac{3}{2x}$$

From (2):

$$4 = 2\lambda y \implies \lambda = \frac{2}{y}$$

Equating the two expressions for λ :

$$\frac{3}{2x} = \frac{2}{y} \implies 3y = 4x \implies y = \frac{4}{3}x$$

Substitute $y = \frac{4}{3}x$ into the constraint $x^2 + y^2 = 4$:

$$x^2 + \left(\frac{4}{3}x\right)^2 = 4$$

$$x^2 + \frac{16}{9}x^2 = 4 \implies \frac{25}{9}x^2 = 4 \implies x^2 = \frac{36}{25} \implies x = \pm \frac{6}{5}$$

For y :

$$y = \frac{4}{3}x = \pm \frac{4}{3} \cdot \frac{6}{5} = \pm \frac{24}{15} = \pm \frac{8}{5}$$

Thus, the points are:

$$\left(\frac{6}{5}, \frac{8}{5}\right) \quad \text{and} \quad \left(-\frac{6}{5}, -\frac{8}{5}\right).$$

Step 4: Evaluate the Objective Function

Evaluate $f(x, y) = 3x + 4y$ at the two points:

1. At $\left(\frac{6}{5}, \frac{8}{5}\right)$:

$$f\left(\frac{6}{5}, \frac{8}{5}\right) = 3\left(\frac{6}{5}\right) + 4\left(\frac{8}{5}\right) = \frac{18}{5} + \frac{32}{5} = \frac{50}{5} = 10$$

2. At $\left(-\frac{6}{5}, -\frac{8}{5}\right)$:

$$f\left(-\frac{6}{5}, -\frac{8}{5}\right) = 3\left(-\frac{6}{5}\right) + 4\left(-\frac{8}{5}\right) = -\frac{18}{5} - \frac{32}{5} = -\frac{50}{5} = -10$$

Step 5: Conclusion

The **maximum value** of $f(x, y) = 3x + 4y$ is:

$$\boxed{10}$$

This occurs at:

$$\boxed{\left(\frac{6}{5}, \frac{8}{5}\right)}.$$

1
3

$$\text{Maximize } Z = 5x + 4y$$

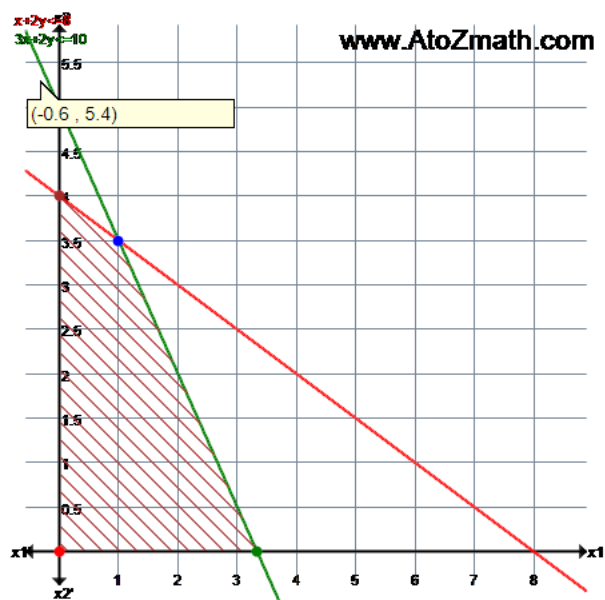
$$x + 2y \leq 8$$

$$3x + 2y \leq 10$$

$$x \geq 0, \quad y \geq 0$$

$$x_1, x_2 \in \mathbb{Z} \quad (\text{integers})$$

Solution



Since $y = 3.5$ is not an integer, we branch by creating two subproblems:

1. Subproblem 1: $y \leq 3$ (i.e., we force y to be less than or equal to 3).
2. Subproblem 2: $y \geq 4$ (i.e., we force y to be greater than or equal to 4).

Subproblem 1: $y \leq 3$

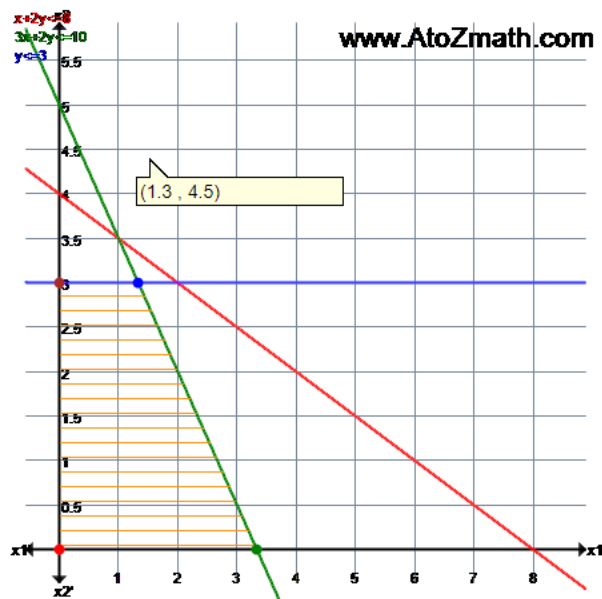
Now, we solve the LP with the additional constraint $y \leq 3$:

Maximize $Z = 5x + 4y$

Subject to:

1. $x + 2y \leq 8$
2. $3x + 2y \leq 10$
3. $y \leq 3$
4. $x \geq 0, \quad y \geq 0$

The graphical soln is:



The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = 5x_1 + 4x_2$
$O(0, 0)$	$4 \rightarrow x_1 \geq 0$ $5 \rightarrow x_2 \geq 0$	$5(0) + 4(0) = 0$
$A(3.33, 0)$	$2 \rightarrow 3x_1 + 2x_2 \leq 10$ $5 \rightarrow x_2 \geq 0$	$5(3.33) + 4(0) = 16.67$
$B(1.33, 3)$	$2 \rightarrow 3x_1 + 2x_2 \leq 10$ $3 \rightarrow x_2 \leq 3$	$5(1.33) + 4(3) = 18.67$
$C(0, 3)$	$3 \rightarrow x_2 \leq 3$ $4 \rightarrow x_1 \geq 0$	$5(0) + 4(3) = 12$

The maximum value of the objective function $Z = 18.67$ occurs at the extreme point $(1.33, 3)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 1.33, x_2 = 3$ and $\max Z = 18.67$.

Subproblem 2: $y \geq 4$

Now, we solve the LP with the additional constraint $y \geq 4$:

Maximize $Z = 5x + 4y$

Subject to:

1. $x + 2y \leq 8$
2. $3x + 2y \leq 10$
3. $y \geq 4$
4. $x \geq 0, y \geq 0$



The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = 5x_1 + 4x_2$
$A(0, 4)$	$1 \rightarrow x_1 + 2x_2 \leq 8$ $3 \rightarrow x_2 \geq 4$	$5(0) + 4(4) = 16$

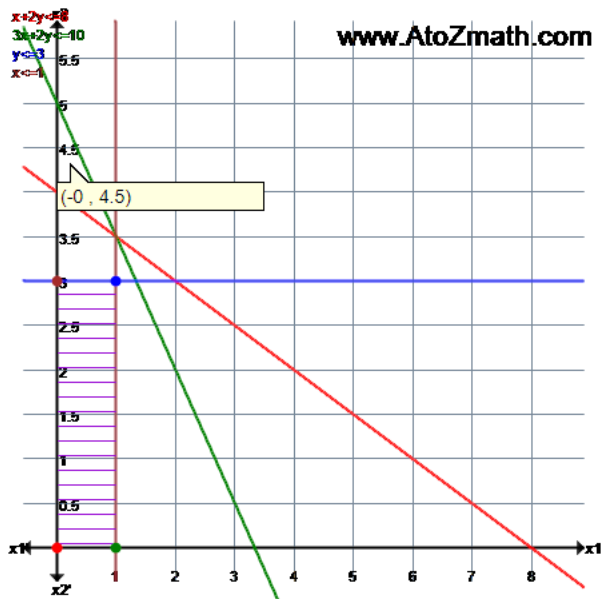
The maximum value of the objective function $Z = 16$ occurs at the extreme point $(0, 4)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 0, x_2 = 4$ and $\max Z = 16$.

This Problem has integer solution, so no further branching is required.

Subproblem 3:

$\text{MAX } Z = 5x_1 + 4x_2$
 subject to
 $x_1 + 2x_2 \leq 8$
 $3x_1 + 2x_2 \leq 10$
 $x_2 \leq 3$
 $x_1 \leq 1$



The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = 5x_1 + 4x_2$
$O(0, 0)$	$5 \rightarrow x_1 \geq 0$ $6 \rightarrow x_2 \geq 0$	$5(0) + 4(0) = 0$
$A(1, 0)$	$4 \rightarrow x_1 \leq 1$ $6 \rightarrow x_2 \geq 0$	$5(1) + 4(0) = 5$
$B(1, 3)$	$3 \rightarrow x_2 \leq 3$ $4 \rightarrow x_1 \leq 1$	$5(1) + 4(3) = 17$
$C(0, 3)$	$3 \rightarrow x_2 \leq 3$ $5 \rightarrow x_1 \geq 0$	$5(0) + 4(3) = 12$

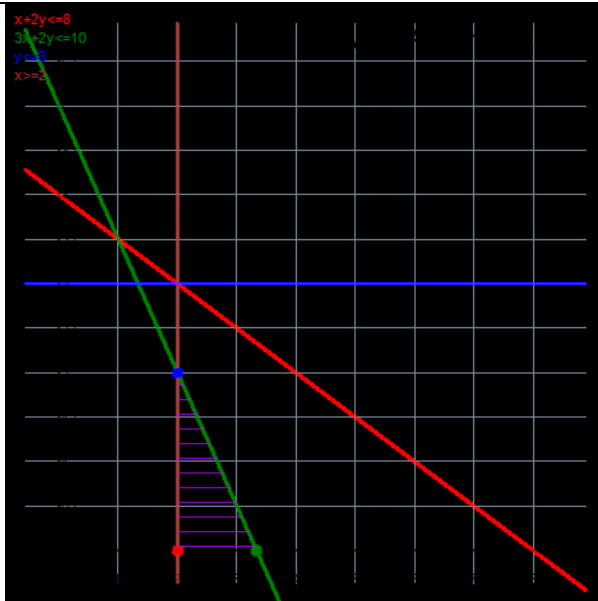
The maximum value of the objective function $Z = 17$ occurs at the extreme point $(1, 3)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 1, x_2 = 3$ and $\max Z = 17$.

This Problem has integer solution, so no further branching is required.

Subproblem 4:

$\text{MAX } Z = 5x_1 + 4x_2$
 subject to
 $x_1 + 2x_2 \leq 8$
 $3x_1 + 2x_2 \leq 10$
 $x_2 \leq 3$
 $x_1 \geq 2$



The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = 5x_1 + 4x_2$
$A(2, 0)$	$4 \rightarrow x_1 \geq 2$ $6 \rightarrow x_2 \geq 0$	$5(2) + 4(0) = 10$
$B(3.33, 0)$	$2 \rightarrow 3x_1 + 2x_2 \leq 10$ $6 \rightarrow x_2 \geq 0$	$5(3.33) + 4(0) = 16.67$
$C(2, 2)$	$2 \rightarrow 3x_1 + 2x_2 \leq 10$ $4 \rightarrow x_1 \geq 2$	$5(2) + 4(2) = 18$

The maximum value of the objective function $Z = 18$ occurs at the extreme point $(2, 2)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 2, x_2 = 2$ and $\max Z = 18$.

This Problem has integer solution, so no further branching is required.

The branch and bound algorithm thus terminated and the optimal integer solution is : $Z=18$ and $x_1=2, x_2=2$

1
4

Max $Z = 100x_1 + 150x_2$
subject to
 $8000x_1 + 4000x_2 \leq 40000$
 $15x_1 + 30x_2 \leq 200$
and $x_1, x_2 \geq 0$

Graphical Solution is

Max $Z = 1055.5556$ with $x_1 = 2.2222, x_2 = 5.5556$.

$x_2 = 5.5556$ must be an integer value, so two new constraints are created, $x_2 \leq 5$ and $x_2 \geq 6$

subproblem 1:

$$\text{Max } Z = 100x_1 + 150x_2$$

subject to

$$8000x_1 + 4000x_2 \leq 40000$$

$$15x_1 + 30x_2 \leq 200$$

$$x_2 \leq 5$$

and $x_1, x_2 \geq 0$;

Solution is

$$\text{Max } Z_B = 1000 \left(x_1 = 2.5, x_2 = 5 \right)$$

Subproblem 2:

$$\text{Max } Z = 100x_1 + 150x_2$$

subject to

$$8000x_1 + 4000x_2 \leq 40000$$

$$15x_1 + 30x_2 \leq 200$$

$$x_2 \geq 6$$

and $x_1, x_2 \geq 0$;

Solution is

$$\text{Max } Z_C = 1033.3333 \left(x_1 = 1.3333, x_2 = 6 \right)$$

In Sub-problem 2, $x_1 (=1.3333)$ must be an integer value, so two new constraints are created, $x_1 \leq 1$ and $x_1 \geq 2$

Subproblem 3.

$$\text{Max } Z = 100x_1 + 150x_2$$

subject to

$$8000x_1 + 4000x_2 \leq 40000$$

$$15x_1 + 30x_2 \leq 200$$

$$x_2 \geq 6$$

$$x_1 \leq 1$$

and $x_1, x_2 \geq 0$;

Solution is

$$\text{Max } Z_D = 1025 \left(x_1 = 1, x_2 = 6.1667 \right)$$

Subproblem 4.

$$\text{Max } Z = 100x_1 + 150x_2$$

subject to

$$8000x_1 + 4000x_2 \leq 40000$$

$$15x_1 + 30x_2 \leq 200$$

$$x_2 \geq 6$$

$$x_1 \geq 2$$

and $x_1, x_2 \geq 0$;

Solution is

This Problem has an infeasible solution, so this branch is terminated.

In Sub-problem 3, $x_2(=6.1667)$ must be an integer value, so two new constraints are created, $x_2 \leq 6$ and $x_2 \geq 7$

Subproblem 5:

$$\text{Max } Z = 100x_1 + 150x_2$$

subject to

$$8000x_1 + 4000x_2 \leq 40000$$

$$15x_1 + 30x_2 \leq 200$$

$$x_2 \geq 6$$

$$x_1 \leq 1$$

$$x_2 \leq 6$$

and $x_1, x_2 \geq 0$;

Solution is

$$\text{Max } Z_F = 1000 \quad (x_1 = 1, x_2 = 6)$$

This Problem has integer solution, so no further branching is required.

Subproblem 6:

$$\text{Max } Z = 100x_1 + 150x_2$$

subject to

$$8000x_1 + 4000x_2 \leq 40000$$

$$15x_1 + 30x_2 \leq 200$$

$$x_2 \geq 6$$

$$x_1 \leq 1$$

$$x_2 \geq 7$$

and $x_1, x_2 \geq 0$;

Solution is

This Problem has an infeasible solution, so this branch is terminated.

The branch and bound algorithm thus terminated and the optimal integer solution is : $Z=1000$ and $x_1=1, x_2=6$